Cloud Computing & Ethics: Discussion and Presentations

Answer every question using your own group project as the case study.

Cloud computing

1. Define cloud computing. Explain its characteristics (on-demand self-service, scalability, elasticity, resource pooling, pay-as-you-go) and show how each would benefit your project.

2. Your group decides to run the project in the cloud. Recommend **one** service model (IaaS, PaaS or SaaS) for training your model and justify your choice. Name one real service that fits (e.g. AWS EC2, Google Colab, Azure Machine Learning, Galaxy).

3. Which deployment model (public, private, hybrid) is most appropriate for your data? Justify it using what your data is and where it came from.

4. Sketch the cloud pipeline for your project: where the data is stored, where preprocessing happens, where the model is trained, and how results reach the end user. Label each stage with the cloud service or component used.

Ethics

1. Research like yours raises ethical questions about data use, the people behind the data and the use of the technology. Explain ethical issues that could arise from your research and suggest how each one could be addressed.

2. Kenya's Data Protection Act 2019 treats genetic and health data as sensitive personal data. If your data were collected from Kenyan patients, state the obligations your group would have.

3. Could your model or results be misused, or cause harm if wrong (e.g. a misdiagnosis, discrimination by an insurer or employer)? Identify one risk and one safeguard.

Ethics checklist

When you meet a case you have never seen, run down this list and pick the two issues the case gives you the most evidence for.

Ask yourself:

Could people or animals in the research suffer pain, stress or harm?

Could it disrupt ecosystems, spread disease or harm native species?

Whose data or samples are used, and did they agree to this use?

Who is missing from the data, and who would get worse results?

Are results announced before peer review, overclaimed or explained poorly?

Should humans be doing this at all, and who decides?

Could the technology or results be used to cause harm?

Does it take money from more urgent work, and who benefits from it?

Practice question: an unseen case (individual)

Ziwa Biotech (a fictional company) is a Nairobi start-up. It has used gene editing to create malaria-carrying mosquitoes that pass a female-sterility gene to almost all of their offspring, a technique called a gene drive. Lab trials show the gene could wipe out a local mosquito population within a few years. The company plans to release the mosquitoes on an island in Lake Victoria next year. It announced the plan at a press conference, describing it as "the end of malaria in East Africa", before publishing any peer-reviewed results. Fishing communities on the island learnt of the plan from the news.

1. Research like Ziwa Biotech's raises ethical questions about ecological impact, community consent and the use of the technology. Explain ethical issues that could arise from this research and suggest how each one could be addressed.